

This question paper contains 2 printed pages!

NEPJD-402-2026

FACULTY OF SCIENCE AND TECHNOLOGY
B.Sc. (Second Year) (Fourth Semester) EXAMINATION

APRIL/MAY, 2026

(NEP-2020)

COMPUTER SCIENCE

Paper-SCSCCT-1252

(Operating System)

(Monday, 13-4-2026)

Time : 2.00 p.m. to 4.00 p.m.

Time—2 Hours

Maximum Marks—30

N.B. — (i) All questions carry equal marks.

(ii) Question No. 1 is compulsory.

(iii) From Q. No. 2 to Q. No. 5, solve any two questions.

1. Answer the following questions (2.5 marks each) : 10

- (a) Explain Extended Machine View.
- (b) Explain Segmentation in Memory Management.
- (c) Explain Round Robin Scheduling.
- (d) Describe Banker's Algorithms.

2. Answer the following questions : 10

- (a) Explain Hierarchical Machine View in Operating System.
- (b) Describe various components of Operating System.

P.T.O.

WT

(2)

NEPJD-402-2026

3. Answer the following questions : 10

(a) Explain internal and external Fragmentation in Memory Management.

(b) Explain Single Contiguous Allocation Techniques.

4. Answer the following questions : 10

(a) Explain Scheduling Criteria in Process Management.

(b) Explain FCFS and SJF Scheduling in Scheduling Algorithms.

5. Answer the following questions : 10

(a) Explain Channels and Control units in Device Management.

(b) Explain a General Model of File Systems.

NEPJD-402-2026

2

X148Y87D417X148Y87D417X148Y87D417X148Y87D417

X148Y87D417X148Y87D417X148Y87D417X148Y87D417